RMSSD (Root Mean Square of Successive Differences)

RMSSD is a time-domain metric of Heart Rate Variability (HRV) that quantifies short-term fluctuations between consecutive heartbeats. It is derived from RR intervals, where each RR interval represents the time between two successive heartbeats.

Calculation

Given a sequence of RR intervals:

$RR_1, RR_2, ..., RR_n$

1. Compute successive differences between RR intervals:
   $\Delta RR_i = RR_{i+1} - RR_i$
2. Square each difference.
3. Compute the mean of the squared differences.
4. Take the square root of the mean:

$$RMSSD = \sqrt{[(1/(N-1)) \cdot \Sigma(\Delta RR_i^2)]}$$

The result is expressed in milliseconds (ms).

Interpretation

RMSSD reflects short-term beat-to-beat variability and is strongly influenced by parasympathetic (vagal) nervous system activity.

- Higher RMSSD values are generally associated with greater parasympathetic activity, relaxation, and physiological recovery.
- Lower RMSSD values are commonly associated with stress, mental workload, fatigue, or reduced vagal activity.

Typical Values (Proposed Ranges)

Based on values reported in the literature for healthy adults at rest, the following ranges are proposed for interpretation in this work:

RMSSD (ms)	Interpretation
< 25 ms	Low HRV, often associated with stress, fatigue, or reduced recovery
25–75 ms	Typical resting range for healthy adults
> 75 ms	High HRV, often associated with strong parasympathetic activity and relaxation

It is important to note that HRV metrics are highly individual, and relative changes with respect to a personal baseline are generally more informative than absolute thresholds.

Heart Rate (HR)

Heart rate (HR) represents the number of heartbeats per minute (bpm). It is directly related to the average RR interval, where shorter RR intervals correspond to higher HR values and longer RR intervals correspond to lower HR values.

$$HR = 60/RR$$

(where RR is expressed in seconds)

Interpretation

HR is influenced by the balance of the autonomic nervous system:

- **Stress** is typically associated with an **increase in HR**, due to sympathetic activation.
- **Sleepiness or relaxation** is generally associated with a **decrease in HR**, due to parasympathetic dominance.

However, HR is considered a relatively **coarse and less specific indicator**, since it reflects overall cardiac rate but does not capture beat-to-beat dynamics in detail.

Relationship with HRV and RMSSD

HR is inversely related to RR intervals and provides a global measure of cardiac activity, whereas HRV metrics such as RMSSD capture the fine-grained variability between consecutive heartbeats.

Sensitivity to Stress

Although both HR and HRV are affected by stress, HR is generally **less sensitive than RMSSD** for detecting autonomic changes.

- HR may increase moderately under stress, but can remain within a narrow range depending on the individual and context.
- RMSSD shows a **strong and consistent decrease under stress conditions**, reflecting reduced parasympathetic activity.

This is supported by physiological evidence showing that HRV metrics, particularly RMSSD, are more sensitive markers of acute stress than HR alone (Task Force, 1996; Kim et al., 2018).

Summary

- **HR** → global measure of cardiac rate (beats per minute), inversely related to RR intervals.
- **RMSSD** → sensitive marker of short-term HRV reflecting parasympathetic activity.
- In stress monitoring, **RMSSD is typically more informative than HR**, while HR provides complementary but less specific information.